

Week 12: Lean metaprogramming

Functional Programming in Lean

Learning goals

- ▶ Separate syntax from elaborated expressions
- ▶ Read a quotation and an antiquotation
- ▶ Define a small macro for terms and tactics

Central idea

A macro transforms parsed syntax into new syntax that Lean then elaborates.

Lean example

```
syntax "twice(" term ")" : term
macro_rules
  | `(twice($x)) => `($x + $x)
#eval twice(21)
```

What to notice

- ▶ Quotation produces syntax data
- ▶ `$x` inserts captured syntax into a quotation
- ▶ The kernel still checks the generated term or proof

Live coding

1. Complete the `twice` macro
2. Add a `close_simple` tactic macro
3. Inspect one successful and one failed expansion

Pause and predict

Does a macro itself prove the theorem it generates?

Exercise connection

Open `exercises/Week12.md` after the lecture.

The complete example is `Course/Week12.lean`; the fill-in version is `Templates/Week12.lean`.